

SECTION A [25 marks]

This section consists of 3 questions. Answer **all** questions.

1 Evaluate

(a) $\lim_{x \rightarrow 2} \sqrt{\frac{x^2 + 5x - 14}{x - 2}}$ [3 marks]

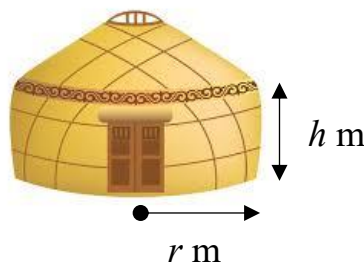
(b) $\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + 2}}$ [3 marks]

(c) $\lim_{x \rightarrow -4^-} \frac{x^3 + 64}{|x + 4|}$ [3 marks]

2 (a) Given $f(x) = \frac{2}{3\sqrt{x}}$. By using the first principle, find $f'(x)$. [4 marks]

(b) By using law of logarithm, show that $y = \ln \sqrt{\frac{1+e^x}{1-e^x}}$ can be written as $y = \frac{1}{2} [\ln(1+e^x) - \ln(1-e^x)]$. Hence, find $\frac{dy}{dx}$ and give your answer in the simplest form. [6 marks]

3 A yurt is built with a hemisphere on top of a cylinder of radius, r m. Given that the total fabric used as the outer layer of the yurt is 216π m².



(a) Show that the volume of the yurt, V m³ is given by $V = 108\pi r - \frac{1}{3}\pi r^3$. [3 marks]

(b) Find the value of r when V is maximum. [3 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer **all** questions.

- 1 Determine the values of x and y that satisfies $\frac{x}{2-i} + yi = (4-3i)2i^2$ where $x, y \in \mathbb{R}$. Hence, state the complex number z in the form $y + xi$. [5 marks]

- 2 (a) Find the solution set for the inequality $\frac{x}{5+x} < \frac{1}{4}$. [5 marks]

- (b) If $\frac{3p^x - p^{-x}}{p^x + 4p^{-x}} = 2$, find x in terms of p . [6 marks]

- 3 (a) Given $f(x) = \sqrt{a + bx}$. Find $f^{-1}(x)$ in terms of a and b where $a, b \in \mathbb{Z}$. Hence, find the value of a such that $f^{-1}(1) = 4$ and $b = -2$. [4 marks]

- (b) Given $f(x) = 5 - e^{-x}$, $x \in \mathbb{R}$ and $g(x) = \ln(x-4)$, $x > 4$. Find $(f \circ g)(x)$. Hence evaluate $\sqrt{(f \circ g)(5)}$. [4 marks]

- 4 Given $f(x) = \frac{2}{x+1}$.

- (a) Find the domain and range of $f(x)$. [2 marks]

- (b) Show that $f(x)$ is a one – to – one function algebraically. [2 marks]

- (c) Determine $f^{-1}(x)$ and its domain. [4 marks]

- (d) Hence, sketch the graph of $f(x)$ and $f^{-1}(x)$ on the same axes. Label your graph completely. [3 marks]

- 5 (a) Find the value of k if $\lim_{x \rightarrow 9} \frac{k}{\sqrt{x-5}} - \frac{2}{\sqrt{x}} = \frac{1}{6}$. [3 marks]

- (b) Given $f(x) = \begin{cases} \frac{4x-4}{2x^2-3x+1} & , x < 1 \\ m^2 - (x+2)^2 & , x = 1 \\ \frac{|1-x|}{x-1} + n & , x > 1 \end{cases}$

where m and n are positive constants.

(i) Find the value of n if $\lim_{x \rightarrow 1} f(x)$ exists. [3 marks]

(ii) If $f(x)$ is continuous at $x = 1$, determine the values of m . [3 marks]

(c) Given $g(x) = \frac{2x}{x^2 - 4}$.

By finding the limit at appropriate values, obtain the vertical and the horizontal asymptotes, if they exist.

[5 marks]

6 (a) Given $(1 + x^2)(1 + y^2) = x^2 - 1$.

(i) Find $\frac{dy}{dx}$ in terms of x and y . [3 marks]

(ii) Hence, show that $(1 + x^2)\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + y = 0$. [2 marks]

(b) Given the parametric equations $x(t) = \sin(2\sqrt{t})$ and $y(t) = \cos^2 \sqrt{t}$ for $t > 0$.

(i) Show that $\frac{dy}{dx} = -\frac{1}{2} \tan(2\sqrt{t})$. [5 marks]

(ii) Hence, evaluate $\frac{d^2y}{dx^2}$ when $t = \frac{\pi}{4}$. Give your answer correct to 1 decimal place. [5 marks]

7 Water is pumped at a constant rate of $500 \text{ cm}^3 \text{ s}^{-1}$ into an empty inverted conical container of radius 400 cm and height 1600 cm.

(a) Show that the rate of change of the height of water in the container is $\frac{dh}{dt} = \frac{8000}{\pi h^2}$ where h is the height of water in the container. [6 marks]

(b) Calculate the rate of change of the radius of water in the container at the instant radius of the surface of water is half of the radius of the container. [4 marks]

END OF QUESTION PAPER